

RelationalRAG-Tables: A Learned Join Verifier for QA over Tables and Knowledge Graphs

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Abstract

Answering questions over relational tables joined with knowledge graphs breaks when identifiers drift across formats, hierarchies, and vocabularies: systems that rely on exact matching fail silently, returning confident wrong answers instead of signaling doubt. We propose **RelationalRAG-Tables**, where an execution-aware planner consults a small learned verifier before committing any join, and a calibrated uncertainty estimate lets the planner abstain when evidence is weak, turning silent errors into explicit refusals with logged verifier scores. The contribution is not a new matching architecture; it is an auditable planner decision and a systematic characterization of when standard learned matching is worth using. We evaluate on a controlled synthetic benchmark (**20** seeds), an ICD-10 to Wikidata table+KG benchmark we construct and run end to end through the planner, value linking and text-to-SQL on BIRD, and evidence retrieval on HybridQA. After correcting BIRD table attribution through SQL-alias resolution, learned verification recovers matches that exact tools miss (five-seed recall on drifted values **0.91** vs. **0.00**) and converts wrong answers into calibrated abstentions: on the constructed table+KG benchmark the gated planner returns 0.7% silently wrong answers against 14.0% ungated. Two conditional theorems characterize when learned verification beats exact matching and when abstention pays; both are checked against measured error rates. We also report the boundaries: hierarchy rules, lexical overlap, and a tuned rule cascade tie or beat the verifier in their regimes, and text-to-SQL gains are bounded by column grounding rather than by

component is a prompted 7B coder; the learned parts are small verifiers plus a fine-tuned MiniLM cross-encoder. The contribution is not a new matching model but a *systematic regime characterization*—backed by calibrated verification and uncertainty-gated abstention as first-class planner primitives—that tells practitioners when learned verification outperforms rules, when it only ties, and when it harms.

Concretely, we deliver: (i) a formulation of QA over (T, G) in which join feasibility is an explicit, learned, uncertainty-gated planner operation (commit, backtrack, or abstain), with two theorems on when learned verification beats exact matching and when abstention pays, both checked against measured error rates over 20 seeds (Theorem 2 predicts the observed join-recall gain to within 0.013; Sections 3 and 6); (ii) a dependency-light core implementation, synthetic generator, and a table+KG benchmark built from real ICD-10-CM identifiers and Wikidata relations with constructed records (Section 4); (iii) a controlled study over 20 seeds with full risk-coverage curves and validation-selected thresholds, showing learned verification lifts EM from $0.580_{\pm 0.012}$ to $0.862_{\pm 0.016}$ under drift while calibrating abstention (Section 5); (iv) real-data evidence at both levels: after SQL-alias correction, at the component level the verifier recovers drifted BIRD values (five-seed recall 0.91 vs. 0.00) and, on the constructed table+KG benchmark run end to end through the planner, cuts silently wrong answers from 14.0% (ungated) to 0.7%; and (v) the boundaries, reported straight: a prefix rule ties the verifier on hierarchy drift, a tuned rule cascade matches it on mixed drift, lexical matching beats it on HybridQA, and in text-to-SQL its hints are bounded by column grounding, quantified with an oracle ceiling. Section 2 surveys related work; Section 6 discusses design rationale and limitations.

2 Related Work

Table and knowledge-graph question answering.

Column-aware Transformers such as TaPas (Herzig et al., 2020) advanced table understanding; semantic parsers such as RAT-SQL (Wang et al., 2020), RESDSQL (H. Li et al., 2023), CodeS (H. Li et al., 2024), and multi-agent extensions (Wang et al., 2025) compile questions into SQL on a single schema and treat joins as engineered JOIN clauses without a learned verifier. KBQA spans skeleton parsers (Sun et al., 2020) and large KBs such as Wikidata; hybrid table-KG QA (Chen et al., 2020) assumes gold cell-to-passage links rather than learning entity identity. Neither line interoperates with relational data under identifier drift; both rely on exact-string entity lookup, the gap our verifier addresses.

Tool-using RAG, selective prediction, and uncertainty.

RAG (Lewis et al., 2020) retrieves text to condition a generator and ReAct (Yao et al., 2023) adds tool calls including SQL engines; we fold *join feasibility* into the planner as a learned, abstention-aware signal rather than treating tool outputs as oracles. Selective prediction (Geifman & El-Yaniv, 2017) trades coverage for reliability, deep ensembles (Lakshminarayanan et al., 2017; Fort et al., 2019) provide calibrated uncertainty, and conformal prediction (Angelopoulos & Bates, 2023) adds distribution-free guarantees;

we integrate ensemble uncertainty into the planner’s commit-or-abstain decision rather than applying it post hoc.

2.1 Entity Matching and Value/Schema Linking

Deciding whether two identifiers from different sources denote the same entity is *entity matching* or record linkage (Konda et al., 2016; Christophides et al., 2020), for which learned and pretrained-LM matchers are now standard (Mudgal et al., 2018; Y. Li et al., 2020). The same operation appears in text-to-SQL as *value* linking (Lin et al., 2020) and *schema* linking (Wang et al., 2020; H. Li et al., 2023, 2024), and in table-to-KB alignment (Efthymiou et al., 2017; Sung et al., 2020).

These literatures are largely separate. Our contribution is not a new matcher but a reproducible study of matching *under identifier drift* across synthetic, BIRD, and ICD-10+Wikidata benchmarks, folded into a retrieval-augmented planner. This connects entity matching, value/schema linking, and structured-data RAG while characterizing when a *learned* matcher outperforms string and rule-based alternatives. Unlike SQL agents (Wang et al., 2025) that orchestrate an LLM over a single schema, we isolate the value-matching step and trace how its errors propagate end-to-end.

Structural matchers for joinable-table discovery decide *which* tables to join; we decide whether two already-retrieved identifiers match. Our difference from Ditto, DeepMatcher, and Magellan (Y. Li et al., 2020; Mudgal et al., 2018; Konda et al., 2016) is not the matcher, whose features and architecture we borrow, but the *decision* it drives: the match probability and its uncertainty are consumed by a planner that commits, backtracks, or abstains, and the value of that decision is measured end to end rather than by pairwise F1. We benchmark against Fellegi-Sunter (Fellegi & Sunter, 1969), a Ditto-style cross-encoder (Y. Li et al., 2020; Mudgal et al., 2018), and train-tuned rule cascades (Sections 6–5.7).

3 Method

We formalize the heterogeneous-QA setting, isolate the `join` operation that our learned verifier targets, and present the system design.

Definition 1 (Heterogeneous QA). *A corpus is $\mathcal{D} = (T, G)$ where $T = \{T_1, \dots, T_M\}$ is a set of relational tables, each T_m a set of rows over typed columns C_m , and $G = (N, E, R)$ is a knowledge graph with node set N , edges $E \subseteq N \times R \times N$, and relation vocabulary R . A question q has a (latent) execution trace $\pi^* = (o_1, \dots, o_L)$ where each o_ℓ is a typed relational operation (*select*, *filter*, *join*, *traverse*, *return*, or *abstain*) and a gold answer y^* (possibly the distinguished value *ABSTAIN*).*

A controller emits operations conditioned on history, and an executor $\mathcal{E}$ maps any prefix to grounded intermediates. We use two metrics. *Exact match* (EM) is $\mathbb{I}[\mathcal{E}(\pi) = y^*]$ after answer normalization, and *faithfulness* asks whether the returned value appears in retrieved cells or edges or is the deterministic result of a traversal. The crux is the `join` operation: given a key cell value ρ and a candidate node ν , we decide whether (ρ, ν) denotes the same entity. Brittle tools implement this as substring equality, whereas we make it a learned predicate.

3.1 Theoretical Properties

We state three properties of the verifier-gated planner that justify the design and are checked empirically.

Theorem 1 (Abstention reduces expected loss under asymmetric costs). *Consider a population of join decisions, scored one decision at a time: a correct accepted join scores $+1$, a wrong accepted join $-\lambda$ ($\lambda > 0$), and an abstention 0 . Assume join decisions are independent and the verifier’s errors are independent of retrieval. For a verifier with false-positive rate α and false-negative rate β on a set of n join decisions with match prevalence π_+ , the expected per-decision utility under gated abstention is*

$$U_{\text{gate}} = \pi_+(1 - \beta) - \lambda(1 - \pi_+)\alpha,$$

while the ungated planner achieves $U_{\text{no}} = \pi_+ - \lambda(1 - \pi_+)$. Gating improves expected utility whenever $\lambda > \pi_+\beta / [(1 - \pi_+)(1 - \alpha)]$, i.e., when wrong joins are sufficiently costly relative to the verifier’s error rates.

Proof. The ungated planner commits every candidate: it accepts correctly on the π_+n true matches and accepts a wrong join on the $(1 - \pi_+)n$ non-matches, so $U_{\text{no}} = \pi_+ - \lambda(1 - \pi_+)$. Under gating, the verifier accepts true positives at rate $(1 - \beta)$ and false positives at rate α , yielding $U_{\text{gate}} = \pi_+(1 - \beta) - \lambda(1 - \pi_+)\alpha$. The difference $U_{\text{gate}} - U_{\text{no}} = -\pi_+\beta + \lambda(1 - \pi_+)(1 - \alpha)$ is positive when $\lambda > \pi_+\beta / [(1 - \pi_+)(1 - \alpha)]$, which is the stated bound. $\square$

Corollary 1 (Wrong-join and abstention rates under gating). *Under the assumptions of Theorem 1, gating reduces the rate of wrong joins from $(1 - \pi_+)$ to $(1 - \pi_+)\alpha$ and introduces an abstention rate of $\pi_+\beta + (1 - \pi_+)(1 - \alpha)$. The guarantee is conditional on α : it bounds wrong admissions only as tightly as the verifier’s measured false-positive rate.*

The condition is measurable. Section 6 reports α , β , π_+ and the implied critical penalty $\lambda^* = \pi_+\beta / [(1 - \pi_+)(1 - \alpha)]$ for every dataset; on the synthetic study $\lambda^* = 0.39$, so gating is predicted to help for any $\lambda \geq 1$, and the sign of this prediction agrees with the planner’s measured utility in 20/20 seeds. At $\lambda=5$ deferring the most uncertain answers raises utility from $0.270_{\pm 0.070}$ to $0.600_{\pm 0.066}$ (Section 5).

Theorem 2 (Regime separation: when learned verification improves over exact matching). *Let $\delta \in [0, 1]$ be the fraction of join pairs whose identifiers are format-drifted (not string-equal to the gold). An exact matcher has recall $\text{Rec}_{\text{ex}} = 1 - \delta$ on matches. A learned verifier with recall r on drifted pairs and recall r_0 on clean pairs achieves $\text{Rec}_{\text{learn}} = (1 - \delta)r_0 + \delta r$. The learned verifier strictly improves recall whenever $\delta > 0$ and $r > (1 - r_0)(1 - \delta)/\delta$, which is satisfied for any $r > 0$ when $r_0 = 1$ (no clean-pair loss) and $\delta > 0$.*

Proof. The improvement $\text{Rec}_{\text{learn}} - \text{Rec}_{\text{ex}} = (1 - \delta)(r_0 - 1) + \delta r$. When $r_0 = 1$ this equals $\delta r > 0$ for any $\delta, r > 0$. Otherwise, the condition is $\delta r > (1 - \delta)(1 - r_0)$, yielding the stated bound. $\square$

The three quantities are measured directly on each corpus’s gold join pairs rather than fitted: over 20 seeds $\delta = 0.553_{\pm 0.021}$, $r = 0.926_{\pm 0.047}$, $r_0 = 0.9996$, giving a

predicted gain of $0.512_{\pm 0.032}$ recall points against an observed join-task EM gain of $0.515_{\pm 0.025}$ (mean absolute error 0.013; Section 6).

Lemma 1 (Ensemble averaging does not worsen the Brier score). *For a K -member ensemble with member predictions $\hat{p}_1, \dots, \hat{p}_K$ and mean $\bar{p} = K^{-1} \sum_k \hat{p}_k$, the Brier score of $\bar{p}$ is at most the mean Brier score of the members, by Jensen’s inequality applied to the convex map $p \mapsto (p - y)^2$. The Brier score decomposes into reliability (calibration) and refinement terms, so the ensemble can trade one for the other; we therefore report calibration empirically rather than claiming a general ECE bound: over 20 seeds the ensemble’s ECE is $0.026_{\pm 0.010}$ against $0.040_{\pm 0.016}$ for a single member, and member correlation (Fort et al., 2019) explains why the K -sweep saturates at $K=3$ (Section 5).*

3.2 System Design

Figure 1 previews the pipeline; cross-source joins are learned, gated operations rather than implicit string matches.

3.2.1 Encoding and Relational Retriever

Each row $r \in T_m$ is encoded with a value embedding plus a column-type embedding $\phi_{\text{col}}(c_j)$, and graph triples symmetrically into per-edge embeddings, using deterministic feature-hash embeddings that require no pretrained model; this removes the confound of memorized facts and isolates the verifier’s contribution (column-aware Transformers (Herzig et al., 2020) are a drop-in extension). We index three views, per-cell $\mathcal{I}_C$, per-column header $\mathcal{I}_H$, and per-edge $\mathcal{I}_E$; given a question q , candidates are scored as $s(q, x) = \langle f_q(q), f_x(x) \rangle / \gamma$ and the top- k of each view are returned.

3.2.2 Execution-Aware Planner

At each step the planner emits a typed operation $o_\ell \in \{\text{select}, \text{filter}, \text{join}, \text{traverse}, \text{return}, \text{abstain}\}$. Every `join` is gated: the planner queries verifier V for a match probability and uncertainty head U for a confidence signal (Algorithm 1). Candidates are tried in retriever-score order; the first clearing τ_v within budget τ_u is committed, rejected candidates trigger backtracking, and an exhausted pool triggers abstention. The planner is a deterministic, rule-guided controller; a learned extension is discussed in Section 6.1.

3.2.3 Learned Join Verifier

Given (ρ, ν) the verifier predicts $\hat{p} = \sigma(g_\phi(\rho, \nu))$. The reference implementation builds an explicit feature vector (exact match, containment, digit agreement, n -gram Jaccard, edit-ratio, length ratio, prefix overlap, embedding cosine, token Jaccard) and trains a small MLP with cross-entropy on positive and hard-negative pairs. Feature ablation confirms the gain is learned fusion: no single feature matches the full model (digit AUROC 0.70, embedding 0.52, full 0.82; supplementary material).

Algorithm 1: RelationalRAG-Tables planner loop

Input: question q , corpus (T, G) , retriever $\mathcal{R}$, verifier V , uncertainty head U , thresholds τ_v, τ_u , step budget $L_{\max}$

```
1  $h_0 \leftarrow \text{encode}(q), \pi \leftarrow [];$ 
2 for  $\ell = 1, \dots, L_{\max}$  do
3    $o_\ell \leftarrow$  next typed operation for  $q$  given history  $h_{\ell-1};$ 
4   if  $o_\ell.\text{type} = \text{join}$  then
5      $\text{committed} \leftarrow \text{false};$ 
6     foreach candidate node  $\nu$  in retriever order do
7        $\hat{p} \leftarrow V(o_\ell.\text{key}, \nu);$ 
8       if  $\hat{p} < \tau_v$  then continue (reject, backtrack);
9       if  $U.\text{uncertainty}(o_\ell.\text{key}, \nu) > \tau_u$  then return ABSTAIN;
10      commit  $\nu$ ,  $\text{committed} \leftarrow \text{true}$ , break;
11   if  $\neg \text{committed}$  then return ABSTAIN;
12    $h_\ell \leftarrow \mathcal{E}(o_\ell, h_{\ell-1}), \pi \leftarrow \pi \cup \{o_\ell\};$ 
13   if  $o_\ell.\text{type} = \text{return}$  then return  $h_\ell;$ 
```

3.2.4 Uncertainty via Deep Ensembles

We train K independently seeded verifiers on bootstrap resamples (Lakshminarayanan et al., 2017) with diverse hidden widths. The ensemble exposes mean probability $\bar{p}$ (gating joins) and predictive uncertainty $H(\bar{p}) = -\bar{p} \log \bar{p} - (1 - \bar{p}) \log(1 - \bar{p})$ (gating abstention); epistemic variance $2 \text{Var}_k(p_k)$ is a diagnostic. In practice we use uncertainty for *selective prediction*—ranking and deferring uncertain cases (Section 5.4).

4 Implementation and Benchmark

The reference implementation requires no external model, GPU, FAISS, or network dependency; the entire study is reproducible from a seed on a laptop. Encoders are deterministic feature-hash embeddings, indices are in-memory JSON, the verifier is a scikit-learn MLP, and the planner is the rule-guided controller of Algorithm 1. This removes the confound of a pretrained model’s memorized facts and isolates the verifier’s contribution.

Synthetic generator

We generate a seeded corpus of an **accounts** table joined to a financial-instruments knowledge graph (instrument $\rightarrow$ asset class $\rightarrow$ regulator), with five task families: single-table lookup, table $\rightarrow$ KG join, multi-hop traverse (two to three hops), bad-join backtrack (a wrong neighboring node precedes the correct one), and abstain (“ghost” accounts absent from the graph). A held-out account split backs the verifier and never appears in QA questions.

Identifier drift

A configurable fraction of KG node identifiers are *format-drifted* relative to the table key by mutating the alphabetic prefix while *preserving the numeric key* (table

ACC-1001 vs. node `account:ACX1001`). This breaks substring and exact matching while leaving the join recoverable by a verifier that weights digit agreement—the same regime as J45.901 vs. J45.9 from the introduction. Distractor nodes are *real neighboring entities*, so wrongly accepting one yields a concrete wrong answer.

Real table+KG benchmark and its end-to-end form

From the 71,704 ICD-10-CM billable codes and the Wikidata disease entities carrying an ICD-10 code (P494) we build a clinical-records table joined to a disease KG (Section 5.7.1). For the end-to-end study (Section 5.7.2) each record drifts by hierarchy, OCR corruption, or vocabulary (a legacy ICD-9 code), 12% of test records are “ghosts” with no KG node, the KG is extended with live-fetched Wikidata relations (specialty, anatomical location, drug, subclass-of, parents’ specialties; 2,126 triples), and the synthetic generator’s five question families are instantiated (518 questions per seed).

Label protocol

Every label is produced by a stated deterministic rule rather than by human claim decomposition. BIRD gold value: the literal of a `col = 'value'` condition in the gold SQL, confirmed in the alias-resolved table and column filtered by that SQL. BIRD drifted: the normalized gold value is not a substring of the normalized question. ICD gold node: the Wikidata node whose ICD-10 code is the category of the record’s billable code, with drift subtype recorded as construction metadata. Two annotators independently audited 100 BIRD and 100 ICD items using the released offline context, followed by joint adjudication of 14 BIRD and 5 ICD rows. On BIRD, Cohen’s κ is 0.81 for binary protocol-gold correctness (the annotators differed on four items, all settled by the gold SQL) and 1.00 for drift; the adjudicated protocol error rates are 0.12 for gold attribution and 0.00 for drift. All 12 errors came from the original extractor choosing the first same-named column containing the literal instead of the alias-resolved table filtered by the SQL. On ICD, both annotators accepted every gold node (error 0.00 and undefined κ because the judgment has no variance) and agreed on drift subtype ($\kappa = 1.00$; error 0.05). Those five errors were no-op OCR substitutions: the valid descendant code remained a hierarchy case, not a corrupted-code case. The self-contained `annotation/` package includes the completed worksheets, adjudications, offline evidence, validation/scoring scripts, and a post-adjudication impact analysis.

Propagation of the audit corrections

The BIRD defect changes the candidate column, so disclosure alone is insufficient. We resolve SQL aliases before column lookup and rerun value linking: the corrected extraction contains 1,103 instances (114 drifted) and changes 89 table assignments. Across five seeds the verifier obtains recall@1 0.939 ± 0.010 overall and 0.905 ± 0.051 on drift, against exact matching at 0.808 ± 0.012 and 0.000; the conclusion survives the correction. The ICD adjudications change only subtype metadata from **corrupt** to **hier**: record codes, descriptions, gold QIDs, pair features, aggregate matching, and end-to-end QA are unchanged. We therefore retain the aggregate ICD results but remove the pre-adjudication subtype breakdown. Section 6 separately reports an

injected training-label-noise sensitivity study; it is a robustness check, not a substitute for these corrections.

Verifier split

Independently, the generator emits training and held-out (key,node) pairs under *graded* corruption: the number of character edits is sampled and the probability that a corrupted identifier denotes the same entity decreases with edit distance, with a $\sim 50/50$ boundary at two edits. This creates genuine label ambiguity, giving the verifier a non-degenerate operating point and the ensemble meaningful predictive variance.

5 Experiments

We first evaluate in a controlled synthetic setting (Sections 5.1–5.6), then validate on real data (Sections 5.7–5.7.4).

5.1 Setup

Unless noted we use a corpus of 600 accounts (seed 7): 300 reserved for the verifier, 60 dev, 36 ghost accounts, and 342 format-drifted nodes at drift fraction 0.6, yielding 1,392 QA questions, 4,000 verifier-training pairs, and 1,500 held-out verifier-evaluation pairs. We compare five join verifiers under the identical planner: **Exact-match** (substring equality, the brittle baseline), **No verifier** (commit the first retrieved candidate, an ablation of the gate), **Learned** (single MLP), **Deep ensemble** ($K=5$), and **Oracle** (the deterministic verifier with access to gold join labels, an upper bound). The join-acceptance threshold is $\tau_v=0.5$. The whole study is repeated over 20 seeds (corpus and verifier seeds vary together) and every number below is a mean over those seeds with the standard deviation typeset as a gray subscript (e.g. $0.862_{\pm 0.016}$) unless stated otherwise; figures show seed 7. One seed runs on a CPU in a few minutes.

5.2 End-to-end QA

The brittle exact-match verifier collapses to $0.580_{\pm 0.012}$ aggregate EM (Figure 2): it cannot recover the 60% of joins whose identifiers are format-drifted (join, multi-hop, and backtrack all fall to 0.45, the undrifted fraction) and consequently *over-abstains*, refusing 620 of 1,392 questions on average. The learned verifier and the ensemble recover the drifted joins (0.95 to 0.96 on join and multi-hop) and reach $0.854_{\pm 0.026}$ and $0.862_{\pm 0.016}$ EM, approaching the oracle ceiling of 1.0 on the easy task families.

Why aggregate EM under-rewards verification

No verifier and the ensemble reach similar aggregate EM ($0.848_{\pm 0.006}$ vs. $0.862_{\pm 0.016}$), but for opposite reasons: the no-verifier planner commits every candidate (1.0 on single/join/multi-hop) yet returns wrong answers for all 36 absent accounts (0.0 on abstain) and for two thirds of the bad-join questions. The learned verifier trades slight join recall for reject/abstain capability. We therefore read verifier value through discrimination, calibration, and selective prediction, and, on real data, through the silent-wrong-answer rate (Section 5.7.2).

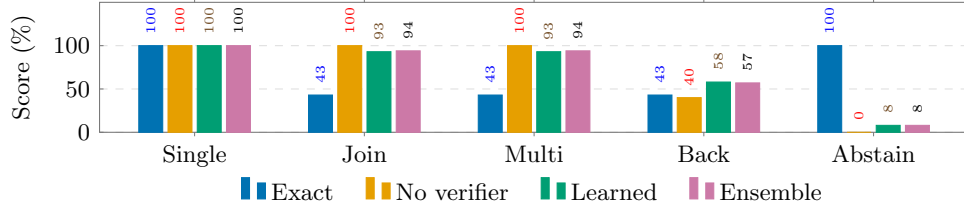


Fig. 2: Synthetic EM by task family (1,392 questions, 60% drift; bars show seed 7). Over 20 seeds the aggregate EM is 0.580 ± 0.012 (exact), 0.848 ± 0.006 (no verifier), 0.854 ± 0.026 (learned), 0.862 ± 0.016 (ensemble), and 1.0 (oracle). Learned verifiers recover drifted joins; exact matching collapses on join-dependent families.

5.3 Held-out verifier discrimination

On 1,500 held-out graded-corruption pairs per seed (Table 1), substring matching has high precision but catastrophic recall (0.37). The learned verifier lifts recall to 0.79 and AUROC to 0.815 ± 0.009 ; the ensemble reaches 0.820 ± 0.008 AUROC with better calibration (Brier $0.37 \rightarrow 0.17$).

Table 1: Held-out join-verifier discrimination on 1,500 graded-corruption pairs unseen during training; mean over 20 seeds, sd as gray subscript.

Verifier	Acc $\uparrow$	Prec $\uparrow$	Rec $\uparrow$	F1 $\uparrow$	AUROC $\uparrow$	Brier $\downarrow$
Exact-match	0.633 ± 0.012	0.961 ± 0.012	0.365 ± 0.014	0.529 ± 0.015	0.673 ± 0.007	0.366 ± 0.012
Learned	0.739 ± 0.015	0.758 ± 0.023	0.789 ± 0.034	0.773 ± 0.016	0.815 ± 0.009	0.174 ± 0.005
Deep ensemble	0.741 ± 0.010	0.760 ± 0.021	0.793 ± 0.038	0.775 ± 0.014	0.820 ± 0.008	0.171 ± 0.004

5.4 Deep-ensemble selective prediction

Figure 3 gives the full risk-coverage curves at 1% coverage steps with 95% bands over seeds, and Table 2 summarizes them. Ranking by predictive uncertainty and deferring the most-uncertain tail raises pairwise accuracy from 0.74 at full coverage to 0.86 at 50% coverage; the area under the risk-coverage curve (AURC, lower is better) is 0.132 ± 0.006 for the ensemble against 0.229 ± 0.011 for exact matching. The ensemble is well calibrated: ECE is 0.026 ± 0.010 , below the single verifier’s 0.040 ± 0.016 and far below exact matching’s 0.37; on BIRD, ECE is 0.001. At the system level, deferring the most-uncertain QA questions raises EM on the answered fraction from 0.88 to 0.95 at 70% and 1.00 at 50% coverage (AURC 0.031 ± 0.007); non-abstained answers are faithful to retrieved evidence in 0.94 of cases.

Thresholds are not tuned on the test split. Selecting τ_v by F1 on a disjoint calibration half gives 0.43 ± 0.06 (range 0.35–0.55 over seeds) with test-half F1 0.779 ± 0.011 , against 0.772 ± 0.016 at the fixed $\tau_v=0.5$ we use throughout; the fixed operating point sits in a flat region. For a target join-risk α , split-conformal selection on the calibration half (Table 2, right) lands close to but slightly above the target on the test half, a finite-sample overshoot we report rather than hide.

When a wrong answer costs $5\times$ a correct one, deferring the most-uncertain answers (optimal coverage $0.62_{\pm 0.08}$) raises utility from $0.270_{\pm 0.070}$ to $0.600_{\pm 0.066}$. The ensemble does not improve selective-prediction *ranking* over a single model (both reach 0.86 at 50% coverage); its value is calibration, consistent with the K -sweep below.

Table 2: Calibration and selective prediction over 20 seeds. *Left:* ECE, MCE, and pairwise AURC. *Right:* split-conformal selection at target risk α (threshold chosen on a calibration half; coverage, risk, and how many of 20 seeds held the target on the test half). *Bottom:* accuracy on the answered fraction at fixed coverages, for held-out pairs and for answered QA questions (system level).

Split	Verifier	ECE ↓	MCE ↓	AURC ↓	α	cov.	risk	held
Synthetic	Exact	$0.366_{\pm 0.012}$	$0.366_{\pm 0.012}$	$0.229_{\pm 0.011}$	0.02	$0.189_{\pm 0.067}$	$0.029_{\pm 0.014}$	4/20
Synthetic	Learned	$0.040_{\pm 0.016}$	$0.082_{\pm 0.033}$	$0.139_{\pm 0.008}$	0.05	$0.273_{\pm 0.036}$	$0.057_{\pm 0.022}$	8/20
Synthetic	Ensemble	$0.026_{\pm 0.010}$	$0.059_{\pm 0.028}$	$0.132_{\pm 0.006}$	0.1	$0.387_{\pm 0.052}$	$0.105_{\pm 0.027}$	10/20
BIRD	Ensemble	0.001	0.053	—				
Accuracy at coverage		1.00	0.90	0.80		0.70	0.50	
Pairs (ensemble)		$0.741_{\pm 0.010}$	$0.770_{\pm 0.012}$	$0.788_{\pm 0.010}$		$0.808_{\pm 0.009}$	$0.858_{\pm 0.009}$	
System (answered)		$0.878_{\pm 0.012}$	$0.900_{\pm 0.014}$	$0.913_{\pm 0.028}$		$0.946_{\pm 0.035}$	$0.999_{\pm 0.003}$	

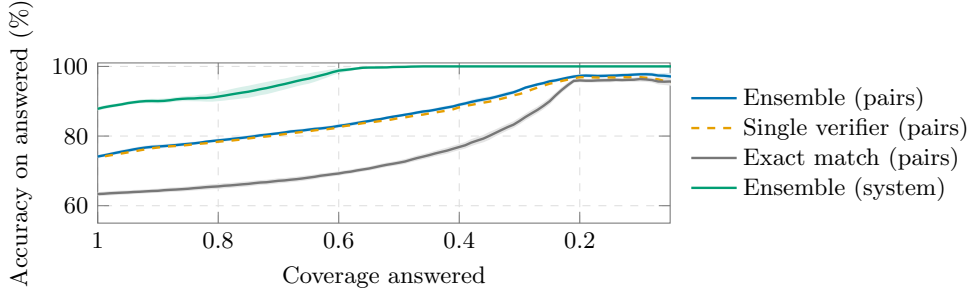


Fig. 3: Full risk-coverage curves (1% coverage steps), mean over 20 seeds with 95% CI bands, for held-out join pairs and for answered QA questions (system level). AURC: exact $0.229_{\pm 0.011}$, single $0.139_{\pm 0.008}$, ensemble $0.132_{\pm 0.006}$ (pairs); $0.031_{\pm 0.007}$ (system).

5.5 Ensemble size and feature ablation

The ensemble’s benefit saturates by $K = 3$: over 20 seeds AUROC is 0.815, 0.820, 0.820 and EM 0.854, 0.862, 0.862 for $K \in \{1, 3, 5\}$ (nested members; per- K selective-prediction rows in the supplementary material). The ensemble buys calibration rather than raw discrimination, and a practitioner who only needs ranking can use a smaller ensemble. Nor is the verifier recovering one obvious rule: retrained on single-feature subsets, digit agreement reaches AUROC 0.70, edit-ratio 0.76, prefix overlap 0.67, and embedding only 0.52, while the full fusion reaches 0.82 (supplementary material). The verifier learns a combination, not a single cue.

5.6 Cost and robustness checks

Per-question latency is ~ 11 ms; verification is sub-millisecond. On an undrifted corpus, exact-match achieves EM 1.0, confirming difficulty stems from drift. A retrieval-noise probe (supplementary material) drops the gold candidate with probability p on the 264 bad-join questions: the no-verifier wrong-answer rate stays at 0.60 for every p , while the ensemble converts failures into abstentions (0.02 at $p=0$ to 0.29 at $p=1$) and at $p=1$ still limits wrong answers to 0.46, a 23% relative reduction in silent errors. The verifier’s gating value *grows* as retrieval quality degrades.

5.7 Real-Data Validation

We now test where the synthetic mechanism transfers and where it does not, at two levels: *component-level* (does the verifier improve matching?) and *end-to-end* (does better matching improve QA?), on a real table+KG benchmark built from ICD-10-CM and Wikidata (Section 5.7.1), on BIRD value linking and text-to-SQL, and on a boundary case (HybridQA) where lexical relevance dominates.

Evidence map.

Table 3 pairs each claim with its strongest result, the level at which it is supported (component-level matching or the complete system), and its statistical status. Boundary results (HybridQA, ICD-10 hierarchy, tuned cascades, text-to-SQL hints) explicitly mark where the verifier only ties, loses, or is bounded by another component.

Table 3: Evidence map: each headline claim, its strongest result, the level at which it is supported, and its statistical status. The corrected BIRD row is produced by the post-adjudication alias-resolution rerun; other rows regenerate from the released harness.

Claim (dataset)	Result	Level	Status
Drifted joins (synth.)	AUROC 0.67 $\rightarrow$ 0.82, EM 0.58 $\rightarrow$ 0.86	pairs, system	20 seeds
Selective pred. (synth.)	ECE 0.026 $\pm$ 0.010, AURC 0.23 $\rightarrow$ 0.13	pairs, system	20 seeds
Theorems 1–2 (synth.)	gain predicted 0.512 $\pm$ 0.032, observed 0.515 $\pm$ 0.025	theory	20 seeds
Value linking (BIRD)	recall@1 0.00 $\rightarrow$ 0.91 on drift	component	5-seed corrected rerun
Hierarchy join (ICD-10)	0.75, ties prefix rule	component	5 seeds, $p=0.59$
Cross-vocab. (ICD-9 $\rightarrow$ 10)	rules 0.01, learned 0.90	component	single seed
Heterogeneous (ICD)	0.75 vs. rule 0.26, tuned cascade ties	component	5 seeds
Table+KG QA (ICD+Wikidata)	wrong 14.0% $\rightarrow$ 0.7%, EM 0.860 vs. 0.839	system	wrong-rate sig. 5/5; EM n.s.
End-to-end EX (BIRD)	oracle hints +3.1, verifier $\leq$ +0.4, pruning+values +0.5	system	full dev, bootstrap
Negative (HybridQA)	lexical 0.25 > verifier 0.09	component	single seed

5.7.1 Real table+KG join: ICD-10 + Wikidata

We build a table+KG benchmark from the 71,704 ICD-10-CM billable codes and 590 Wikidata disease entities carrying an ICD-10 code (property P494). KG nodes carry category-level codes (e.g. R50); table records carry specific billable codes (e.g. R509). Of the 590 records, 164 (28%) are drifted, and for 159 of them *no* KG node carries the record’s exact code.

Evaluation protocol.

The ICD candidate pool is all 590 Wikidata disease nodes; for BIRD, the true column domain. We shuffle before scoring; train/test split on records with only relational pair features (supplementary material).

On real ICD-10 identifiers, **exact matching scores 0.000 on drift** (consolidated baselines in the supplementary material). The learned verifier recovers those joins at 0.746 (overall $0.64 \rightarrow 0.83$, $+0.18$, paired bootstrap $p < 10^{-3}$). A prefix-hierarchy rule ties the verifier (0.826 vs. 0.826 overall, $p=0.59$); canonicalization leaves exact at 0.000 on drift, confirming genuine hierarchy drift. Jaro–Winkler (Winkler, 1990) ties the prefix rule (0.729 on drift) and character TF-IDF (Sung et al., 2020) trails (0.407). Under OCR/typo corruption the prefix rule falls ($0.73 \rightarrow 0.40$) while the verifier stays competitive, though it does not significantly beat the best regime-specific baseline (two-way mix: 0.58 vs. 0.59, $p=0.86$; supplementary material). The takeaway: on deterministic hierarchical identifiers a structural rule is near-optimal; learning wins when structure breaks. Results are stable across five seeds ($0.759_{\pm 0.027}$ vs. prefix $0.791_{\pm 0.044}$).

Cross-vocabulary.

Joining ICD-9 records to the ICD-10 KG (322 diseases, 0% prefix overlap): exact/fuzzy/prefix all score ≈ 0.01 , while embedding reaches 0.91 and the verifier 0.90. Rules win when hierarchy is explicit; semantic matching is required across vocabularies.

Heterogeneous drift.

A clinical column can mix hierarchy-coded, OCR-corrupted, and legacy ICD-9 codes. We build this benchmark by drawing one available mechanism per record. No fixed *single* baseline covers the resulting mixture: prefix scores 0.258, embedding 0.689, while a single learned verifier scores **0.752** ($+0.49$ over prefix, $p < 10^{-3}$ in all five seeds; $+0.06$ over embedding, significant in 3 of 5). The label audit found that some attempted OCR edits were no-ops; relabeling those records changes only the subtype metadata, not any code, gold node, feature, prediction, or aggregate score. We consequently make no subtype-specific performance claim.

Can hand-tuned fusion match it?

Yes. A three-stage cascade whose order and thresholds are tuned on the training split (prefix if it fires, else fuzzy at ≥ 0.9 , else embedding) reaches $0.776_{\pm 0.021}$ and matches the verifier (difference -0.019 , not significant in 4 of 5 seeds), and so does a logistic regression on the same six features (supplementary material). Pool-relative features

(each score minus the pool maximum, the information a cascade exploits) halve the verifier’s seed variance without moving its mean. Any fusion that sees all three signals lands at 0.76–0.78: the learned verifier’s advantage is not capacity but a calibrated probability the planner can gate and abstain on (Section 5.7.2), which a cascade does not emit.

5.7.2 End-to-end QA on the real table+KG benchmark

The ICD results above are component-level. We now run the complete framework on the same real identifiers and Wikidata relations, with constructed clinical records under heterogeneous drift and the paper’s planner over five question families (Section 4), 518 questions per seed, five seeds. Only the gate changes between rows of Table 4.

Table 4: End-to-end QA on the real ICD-10 + Wikidata table+KG benchmark under the identical planner (five seeds). “Wrong” is the silent-wrong-answer rate, “false abst.” abstention on answerable questions; utility scores correct +1, wrong $-\lambda$, abstain 0.

Gate	EM $\uparrow$	wrong $\downarrow$	false abst. $\downarrow$	$U_{\lambda=1}$	$U_{\lambda=2}$	$U_{\lambda=5}$
Exact code match	0.354 $\pm$ 0.007	0.001	0.645	0.354	0.353	0.351
Prefix / hierarchy rule	0.493 $\pm$ 0.014	0.003	0.505	0.490	0.487	0.479
Embedding cosine ($\tau=0.62$)	0.885 $\pm$ 0.021	0.111	0.004	0.775	0.664	0.332
No verifier	0.860 $\pm$ 0.019	0.140	0.000	0.720	0.579	0.159
Learned verifier	0.839 $\pm$ 0.015	0.007	0.154	0.832	0.825	0.804
Oracle (UB)	0.935 $\pm$ 0.021	0.000	0.065	0.935	0.935	0.935

Exact and prefix gates abstain on half or more of the questions because their recall collapses on OCR and cross-vocabulary records. The ungated planner and an embedding gate reach the highest EM (0.860 and 0.885) but return a silently wrong answer on 14.0% and 11.1% of questions and score 0.0 and 0.25 on the abstain family. The learned verifier gate answers with EM 0.839 (within 2.1 points of ungated, not significant in any seed) while cutting silent wrong answers to 0.7%; its committed joins are 99.1% correct (system-level AURC 0.004), and it beats the exact and prefix gates in 5/5 seeds. At every evaluated penalty $\lambda \in \{1, 2, 5\}$, the gated framework has the highest utility of all non-oracle gates (0.804 vs. 0.159 ungated at $\lambda=5$). This is the paper’s claim, “convert silent errors into explicit refusals”, demonstrated by the complete system on real identifiers; the residual gap to the oracle (0.935) is false abstention on hard OCR and sibling-code joins.

5.7.3 BIRD value linking

BIRD (J. Li et al., 2023) is a text-to-SQL benchmark over 95 real databases with frequent value drift (e.g. “high school” $\rightarrow$ HS). After resolving SQL aliases before identifying the filtered table, we extract 1,103 **value-linking instances** from the 1,534 dev questions, of which 10.3% **are drifted**. The learned verifier (MLP ensemble over identifier features plus `bge-small` cosine) is trained on a split and evaluated on a held-out split.

Across five corrected runs, exact matching reaches 0.808 ± 0.012 overall but 0.000 **on drift**; fuzzy reaches 0.167 ± 0.038 on drift and embedding 0.806 ± 0.031 . The learned verifier reaches 0.939 ± 0.010 **overall and** 0.905 ± 0.051 **on drift** (AUROC 0.996 ± 0.002). Thus the audit correction changes the candidate domains and individual baselines but not the central result that learned verification recovers drifted values that exact matching cannot. The previously reported cross-encoder used the original extractor and is retained only as an exploratory capacity check in the supplement, not as corrected headline evidence.

5.7.4 End-to-end execution accuracy

We run full text-to-SQL with a local Qwen2.5-Coder-7B-Instruct (4-bit), scoring Execution Accuracy (EX) on BIRD dev. The base pipeline reaches EX 0.458 on 800 questions (simple 0.53, moderate 0.35, challenging 0.37). Value linking helps only after schema linking focuses the prompt: naively injecting hints reaches 0.446, below baseline, so the bottleneck is *schema linking* (H. Li et al., 2023, 2024; Wang et al., 2025). With a learned table selector (recall 0.93), at full-dev scale ($n=1534$, 131 drift) EX rises $0.454 \rightarrow 0.477$ (+2.3, $p=0.008$, CI [+0.5, +4.2]), drift EX $0.33 \rightarrow 0.42$ (+9.2, $p=0.005$), and execution errors fall $0.164 \rightarrow 0.143$ (paired bootstrap, 10,000 resamples); a 14B coder shows the same +2.3 ($n=400$). Embedding-only pruning hurts (0.473 vs. 0.476 full schema), learned pruning helps (0.483), and an oracle at 0.525 shows large schema-linking headroom (supplementary material).

Isolating the verifier end to end.

To attribute the gain, we hold the pipeline fixed at pruning + values and vary only the value linker that fills the hint slot (Table 5). All linkers share the same candidate pool, column-relevance gate, three-hint cap, and prompt slot; the learned verifier is cross-fitted over two question folds. Every row was run with the official AWQ 4-bit coder on a CUDA server (the submission used an MLX coder); the baseline reproduces (EX 0.460 vs. 0.454; drift 0.344 vs. 0.328). These expensive generations characterize the submitted, schema-order extractor. The post-hoc alias correction changes its cross-fit training pools, so we do not present E1 as a corrected re-estimate; EX itself remains valid because predictions are scored directly against BIRD gold SQL. The corrected component result above and the ICD end-to-end experiment carry the post-adjudication claims.

Attributing the gain.

Three findings follow for the submitted pipeline. (i) Value linking has end-to-end headroom: gold literals in the same advisory slot lift EX by +3.1 overall and +9.9 on drift, and drifted literals alone by +6.1 on drift. (ii) The learned verifier is the best non-oracle linker in this run but does not reach that headroom: its hint precision (0.54–0.70) is four times that of exact or fuzzy linking, both of which *hurt* EX by about six points, and its effect turns positive only at a strict threshold ($\tau=0.8$, +0.4); its confidence is still informative, since EX on hinted questions rises from 0.495 to 0.577 when the least-confident half is deferred. (iii) Column grounding limits the result: the corrected verifier reaches five-seed recall@1 0.91 on drifted values when the column

Table 5: Verifier-isolated end-to-end ablation on full BIRD dev ($n=1534$, 131 drifted): schema pruning + column values fixed, only the value linker varied. “hinted q ” is the number of questions that received a hint (hint precision in parentheses); Δ in EX points vs. the no-hint row, paired bootstrap (10^4 resamples).

Value linker	EX	EX drift	hinted q (prec.)	Δ EX [95% CI]	Δ EX, drift [95% CI]
no pruning, no values	0.460	0.344	0	-0.5 [-2.4,1.4]	-5.3 [-13.0,2.3]
none (pruning + values)	0.465	0.397	0	—	—
exact	0.406	0.328	1210 (0.14)	-5.9 [-7.6,-4.2]	-6.9 [-13.0,-1.5]
fuzzy	0.404	0.321	1412 (0.12)	-6.1 [-7.9,-4.2]	-7.6 [-13.0,-3.0]
embedding	0.451	0.359	808 (0.26)	-1.4 [-2.8,-0.1]	-3.8 [-10.7,2.3]
verifier, $\tau=0.5$	0.460	0.359	549 (0.54)	-0.5 [-1.5,0.5]	-3.8 [-8.4,0.0]
verifier, $\tau=0.8$	0.469	0.405	321 (0.63)	+0.4 [-0.3,1.1]	+0.8 [-3.8,5.3]
oracle gold literals	0.495	0.496	828 (0.98)	+3.1 [1.9,4.2]	+9.9 [4.6,16.0]
oracle, drifted only	0.469	0.458	107 (1.00)	+0.5 [0.0,0.9]	+6.1 [0.8,11.5]

is known, but end to end it must also decide *which* column and *whether* a value is referenced. The verifier’s corrected evidence is component-level on BIRD and system-level where the join is an explicit, gated planner operation (Section 5.7.2). We also note that the pruning + values gain itself is backend-sensitive: +2.3 EX ($p=0.008$) with the MLX coder in the submission, 0.465 vs. 0.460 (+0.5, not significant) with the AWQ coder.

5.7.5 Negative result and regime boundaries

In **HybridQA** (Chen et al., 2020), cell-to-passage links are gold-given, so the task is lexical evidence retrieval rather than entity matching: lexical overlap beats dense retrieval (recall@20 0.83 vs. 0.65), the identifier-trained verifier loses at top-1 (0.09 vs. 0.25), and stronger retrievers or a fine-tuned cross-encoder (0.21) do not close the gap (supplementary material). A cascade cannot win both regimes (supplementary material): cheap-primary matches lexical on HybridQA (0.25) but regresses on BIRD; verifier-primary preserves BIRD but only partially recovers HybridQA (0.15), because two separately trained matchers score on incompatible scales (the BIRD verifier’s ECE on HybridQA pairs is 0.117). Training *one* verifier on the pooled pairs of both datasets with features computable in both regimes plus pool-relative and regime features improves cross-regime calibration and lifts HybridQA top-1 from $0.146_{\pm 0.043}$ to $0.237_{\pm 0.030}$, two thirds of the way to lexical ($0.281_{\pm 0.030}$; three seeds, supplementary material). This E4 run predates the BIRD alias correction and is therefore an exploratory routing result, not part of the corrected BIRD headline. Beating lexical ranking on lexical tasks remains open.

6 Discussion

Operating point and calibration. Accept-decision F1 stays within [0.75, 0.77] (synthetic) and [0.79, 0.80] (real ICD pairs) for $\tau_v \in [0.4, 0.6]$, while a validation-selected τ_v changes test F1 by only 0.007 (Section 5.4). Risk-controlled conformal selection (Geifman & El-Yaniv, 2017; Angelopoulos & Bates, 2023) is therefore a complementary layer rather than a competitor. Correlated members yield an effective ensemble

size below K (Fort et al., 2019; Lakshminarayanan et al., 2017), yet post-hoc recalibration of a single model (temperature 0.039, isotonic 0.043) still does not reach the ensemble’s ECE of 0.024.

Theory check. Theorems 1 and 2 are conditional on measurable verifier quantities, which we measure directly (supplementary material). On the synthetic benchmark, $\pi_+ = 0.56$, $\alpha = 0.33$ at $\tau_v=0.5$ (the graded-corruption split is ambiguous by design), $\beta = 0.21$, $\lambda^* = 0.39$. On the alias-corrected BIRD seed-0 split, $\pi_+ = 0.023$, $\alpha = 0.0046$, $\beta = 0.054$, and $\lambda^* = 0.0013$. The theorem therefore predicts positive utility at the evaluated $\lambda \in \{1, 2, 5\}$; on synthetic QA the sign agrees with measured planner utility in every seed. This is not an unconditional safety guarantee. The corollary’s *magnitude* is optimistic (observed QA wrong-rate ratio $0.77_{\pm 0.07}$ vs. $\alpha = 0.33$) because QA-time distractors are real neighbouring nodes, harder than the pairs on which α is measured. Theorem 2 transfers in size on the controlled benchmark (predicted $0.512_{\pm 0.032}$, observed $0.515_{\pm 0.025}$).

Label noise. The audit corrections are propagated directly in Section 4; the following injected-noise sweep is only a sensitivity check. Corrupting a fraction ρ of *training* gold assignments (on ICD, half of them same-category siblings) while scoring the clean test split changes BIRD recall@1 from 0.928/0.947 (all/drift) to 0.910/0.895 at $\rho=0.2$, and ICD recall from 0.752 to 0.733. Across the evaluated levels, FPR remains below 0.47% on BIRD and 0.09% on ICD, while BIRD FNR reaches 25.6% and ICD FNR 34.1% at $\rho=0.2$. Thus this particular perturbation primarily increases rejection rather than false admission at high noise; it does not establish that arbitrary labeling errors can never produce silent wrong joins.

Advisory hints versus gated joins. In the table+KG planner (Section 5.7.2) the join is an explicit operation the verifier decides, with abstention as a first-class answer, which is why gating removes 13 points of silent wrong answers. In text-to-SQL (Table 5) the verifier’s output is an advisory hint that a prompted coder may ignore, the linker must also ground the column, and only a strict threshold yields a small gain. The verifier earns its place where the join is a decision, not a suggestion.

Failure analysis and where gating harms. When the KG node is absent the verifier abstains: removing 10% to 30% of KG edges lowers EM from 0.94 to 0.52 while the wrong-answer rate stays under 0.05 (supplementary material). On the alias-corrected BIRD seed-0 split at $\tau_v=0.5$, the false-positive rate is 0.46% against a false-negative rate of 5.43%. On the ICD-10 join, 18 of 41 recall@1 misses are *sibling-code confusions* that the verifier mostly commits (38 of 41) rather than abstains on. Gating is counter-productive when lexical matching suffices (HybridQA), where it also rejects 6% of correct positives on clean joins (EM $1.00 \rightarrow 0.94$).

Further checks (qualitative examples, latency and scaling, LLM matchers, noise robustness, a Fellegi–Sunter baseline at AUROC 0.753 vs. 0.820) are in the supplementary material.

6.1 Limitations and Future Work

End-to-end scope. The BIRD pipeline uses a local 7B coder (EX 0.45 to 0.48), below cloud-scale systems. Relative gains are backend-sensitive (+2.3 EX with the MLX

coder and a 14B model, +0.5 and not significant with the AWQ coder), while the verifier’s hints are bounded by column grounding (Table 5), which makes a column-level schema linker and cloud-scale backbones the natural next step. The expensive BIRD generations report the submitted extractor; after the audit we reran the tractable component experiment but not the LLM generations, whose EX labels remain the gold SQL. The table+KG study uses a rule-guided planner over a 545-node KG, leaving larger graphs to future work.

Idealized retrieval, rule-guided planner, semi-real benchmark. The synthetic candidate pool usually contains the gold node, which flatters the no-verifier baseline, while the verifier’s advantage grows as retrieval degrades. The controller is deterministic, with a verifier-aware trained planner as future work. The ICD benchmarks use real codes and Wikidata relations over constructed records, where SNOMED CT and RxNorm would broaden validity.

Statistical methodology. Synthetic results are 20-seed means with sd, t -based 95% CIs, and bootstrap bands on risk–coverage curves; designated real-data results use 5-seed means and paired bootstrap p -values (10,000 resamples). The heterogeneous-ICD gain over embedding (+0.06) is significant in 3/5 seeds, where a tuned cascade matches the verifier. The end-to-end table+KG EM gap between gated and ungated planners is not significant in any seed, whereas the silent-wrong-answer gap is. Conformal selection slightly overshoots its target risk in most seeds (Table 2). Two independent annotators audited the deterministic label rules on 100 BIRD and 100 ICD items ($\kappa = 0.81$ for BIRD gold correctness, 1.00 for both drift labels, adjudicated protocol error rates 0.12/0.00 and 0.00/0.05, Section 4); the BIRD component was rerun after correction, and uncorrected ICD subtype statistics are not cited.

Schema and drift scope. The generator joins one table to one graph. Composite keys are validated (AUROC 0.93/0.89), whereas multi-table joins, group-bys, set operations, and temporal drift are out of scope, the verifier scores pairwise joins only, and the synthetic drift model may not match real key formats.

Reproducibility. Every number is tied to a machine-readable output. The synthetic core needs only scientific Python (one seed in minutes on a CPU), while the real-data layer needs a sentence-embedding model and, for text-to-SQL, a local quantized 7 to 14B coder. Runs are seeded, tables are rendered from outputs, the reproduction guide maps each table and figure to its command, and the annotation package provides the alias-corrected BIRD rerun command and result snapshot.

7 Conclusion

Reliable QA over tables and knowledge graphs turns on a decision most systems leave implicit: whether two identifiers from different sources name the same entity. We have argued that this decision should be a first-class, learned, calibrated planner operation. Rather than commit a silent wrong join, RelationalRAG-Tables abstains when unsure. After correcting table attribution through SQL-alias resolution, learned verification recovers matches exact tools miss entirely (0.91 ± 0.05 vs. 0.00 on drifted BIRD values over five seeds). It ties rules and tuned cascades where structure is regular, then yields to lexical matching where relevance is lexical. Two conditional theorems say when

it should help, with predictions checked against measured error rates. End to end, the verifier pays off where the join is a decision the planner owns: on a table+KG benchmark built from real identifiers and Wikidata relations with constructed records it cuts silently wrong answers from 14% to under 1%. Where it is only an advisory hint to a SQL coder, column grounding bounds its payoff, a bottleneck we name precisely. The accompanying primitive, benchmarks, harness, and audit package let others test these boundaries.

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- **Ethics approval, consent to participate, and consent for publication:** Not applicable.
- **Data and code availability:** BIRD, HybridQA, ICD-10-CM, and Wikidata are public. The core harness, constructed ICD-10 + Wikidata benchmarks, pinned splits, and original experiment outputs are available under the MIT license at <https://github.com/lxbachcs/relationalrag-tables>. The revised submission package additionally contains the completed annotation audit, alias-correction driver and corrected snapshot used for the revised BIRD rows; these files will be synchronized to the public repository before deposit. An archived, DOI-bearing release will accompany the camera-ready version.
- **Declaration of generative AI use:** The authors used an AI assistant (Claude, Anthropic) to help write experiment drivers and edit the text. The authors reviewed and edited all content, ran and checked every experiment themselves, and take full responsibility for the publication. No generative AI was used to produce data, results, or figures.
- **Author contributions:** All authors contributed to the study conception and design. X.-B. Le developed the system, ran experiments, and drafted the manuscript. P.T. Tran Truong contributed to data curation and evaluation scripts. H.X. Son supervised the project and revised the manuscript. All authors read and approved the final manuscript.

References

- Angelopoulos, A.N., & Bates, S. 2023. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 164, 494–591. <https://doi.org/10.1561/22000000101>
- Chen, W., Zha, H., Chen, Z., Xiong, W., Wang, H., & Wang, W.Y. 2020. HybridQA: A dataset of multi-hop question answering over tabular and textual data. In *Findings of emnlp*. <https://doi.org/10.18653/v1/2020.findings-emnlp.91>
- Christophides, V., Efthymiou, V., Palpanas, T., Papadakis, G., & Stefanidis, K. 2020. An overview of end-to-end entity resolution for big data. *ACM Computing Surveys*, 536, . <https://doi.org/10.1145/3418896>

- Efthymiou, V., Hassanzadeh, O., Rodriguez-Muro, M., & Christophides, V. 2017. Matching web tables with knowledge base entities: From entity lookups to entity embeddings. In *International semantic web conference (iswc)*. https://doi.org/10.1007/978-3-319-68288-4_16
- Fellegi, I.P., & Sunter, A.B. 1969. A theory for record linkage. *Journal of the American Statistical Association*, 64:328, 1183–1210. <https://doi.org/10.1080/01621459.1969.10501049>
- Fort, S., Hu, H., & Lakshminarayanan, B. 2019. *Deep ensembles: A loss landscape perspective*. <https://arxiv.org/abs/1912.02757>
- Geifman, Y., & El-Yaniv, R. 2017. Selective classification for deep neural networks. In *Advances in neural information processing systems (neurips)*. https://proceedings.neurips.cc/paper_files/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html
- Herzig, J., Nowak, P.K., Müller, T., Piccinno, F., & Eisenschlos, J.M. 2020. TaPas: Weakly supervised table parsing via pre-training. In *Proceedings of the 58th annual meeting of the association for computational linguistics (acl)*. <https://doi.org/10.18653/v1/2020.acl-main.398>
- Konda, P., Das, S., Suganthan G. C., P., Doan, A., Ardalan, A., Ballard, J.R., Raghavendra, V. 2016. Magellan: Toward building entity matching management systems over data science stacks. *Proceedings of the VLDB Endowment*, 9:13, 1581–1584. <https://doi.org/10.14778/3007263.3007314>
- Lakshminarayanan, B., Pritzel, A., & Blundell, C. 2017. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Neurips*. <https://arxiv.org/abs/1612.01474>
- Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Kiela, D. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in neural information processing systems (neurips)* (Vol. 33). <https://proceedings.neurips.cc/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html>
- Li, H., Zhang, J., Li, C., & Chen, H. 2023. RESDSQL: Decoupling schema linking and skeleton parsing for text-to-SQL. In *Aaai*. <https://doi.org/10.1609/aaai.v37i11.26535>
- Li, H., Zhang, J., Liu, H., Fan, J., Zhang, X., Zhu, J., Chen, H. 2024. CodeS: Towards building open-source language models for text-to-SQL. *Proceedings of the ACM on Management of Data*, 2:3, 1–28. <https://doi.org/10.1145/3654930>
- Li, J., Hui, B., Qu, G., Yang, J., Li, B., Li, B., Li, Y. 2023. Can LLM already serve as a database interface? a BIG bench for large-scale database grounded text-to-SQLs. In *Neurips datasets and benchmarks*. <https://doi.org/10.48550/arXiv.2305.03111>
- Li, Y., Li, J., Suhara, Y., Doan, A., & Tan, W.-C. 2020. Deep entity matching with pre-trained language models. *Proceedings of the VLDB Endowment*, 14:1, 50–60. <https://doi.org/10.14778/3421424.3421431>
- Lin, X.V., Socher, R., & Xiong, C. 2020. Bridging textual and tabular data for cross-domain text-to-SQL semantic parsing. In *Findings of emnlp*. <https://doi.org/10.18653/v1/2020.findings-emnlp.438>
- Mudgal, S., Li, H., Rekatsinas, T., Doan, A., Park, Y., Krishnan, G., Raghavendra, V. 2018. Deep learning for entity matching: A design space exploration. In *Sigmod*. <https://doi.org/10.1145/3183713.3196926>
- Sun, Y., Zhang, L., Cheng, G., & Qu, Y. 2020. SPARQA: Skeleton-based semantic parsing for complex questions over knowledge bases. In *Aaai*. <https://doi.org/10.1609/aaai.v34i05.6426>
- Sung, M., Jeon, H., Lee, J., & Kang, J. 2020. Biomedical entity representations with synonym marginalization. In *Acl*. <https://doi.org/10.18653/v1/2020.acl-main.335>
- Wang, B., Ren, C., Yang, J., Liang, X., Bai, J., Chai, L., Li, Z. 2025. MAC-SQL: A multi-agent collaborative framework for text-to-SQL. In *Proceedings of the 31st international conference on computational linguistics (coling)* (pp. 540–557). <https://aclanthology.org/2025.coling-main.36/>
- Wang, B., Shin, R., Liu, X., Polozov, O., & Richardson, M. 2020. RAT-SQL: Relation-aware schema encoding and linking for text-to-SQL parsers. In *Acl*. <https://doi.org/10.18653/v1/2020.acl-main.677>
- Winkler, W.E. 1990. String comparator metrics and enhanced decision rules in the fellegi-sunter model of record linkage. In *Proceedings of the section on survey research methods, american statistical association* (pp. 354–359). <https://eric.ed.gov/?id=ED325505>
- Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. 2023. ReAct: Synergizing reasoning and acting in language models. In *International conference on learning representations (iclr)*. <https://arxiv.org/abs/2210.03629>